

SYNOPSIS
OF
Study of Contraction Mapping Theorems in Metric and
Multiplicative metric Spaces



A Synopsis of the Proposed Research Plan
Submitted for the Degree of
Doctor of Philosophy in

MATHEMATICS

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5. **Proposed plane of research work**

1. We shall try to improve and generalize the results in multiplicative metric spaces and fixed point theorems for contractive mappings.
2. We shall try to extend and generalize the results on common fixed point for in integral type Contractive maps in multiplicative metric spaces.
3. We shall try to generalize the results of fixed point and common fixed point theorems in Cone metric spaces, Multiplicative cone metric spaces, multiplicative cone b- metric spaces.

6. **Conclusion/ Results / Expected out comes of research work**

We shall find some good results based on existence and uniqueness of fixed point theorems and generalize the existing various contraction conditions as well as establish some new contraction condition to extend several fixed point theorems in metric and multiplicative metric spaces.

References

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common fixed point theorem using the idea of compatibility together with conditionally reciprocal continuity of self-mapping in multiplicative metric spaces. In the same year, Khan, A. et al. (2018), found fixed point theorems on multiplicative metric space by the help of integral type contraction of self quadruple mappings. In 2018, Rashid, M.H.M., introduces the concept of cone multiplicative metric space and proved some fixed point results in cone multiplicative metric space.

Just recently, Sharma, R. and Thakur, D. (2020), established some common fixed point theorems for two pairs of self-mappings by using the notion of compatibility of type (E) and weak sub sequential continuity in multiplicative metric spaces. Additionally, Monika and Hooda (2020) gained some fixed point theorems in multiplicative metric space using Kannan contraction mappings.

3. Objective

In specific term, the objective of this study as follows:

1. To study the existence of fixed points and extend various known results to various spaces
2. To modify the fixed point theorem in various abstract space such as metric space, partial metric space, cone metric space, multiplicative metric spaces, etc. to introduced suitable theorem.
3. To apply the functional analysis techniques in certain developed spaces, mappings and iterating process.
4. To examine the pure and applied part of functional analysis.

4. Methodology

We propose the following methodology during the title of research work:

1. To use on line tools, such as google, cite seer, ACM digital library etc. for information and to see present status of concerned research problems.
2. To collect reprints of research papers required for our research work.
3. To attend the seminars / Conferences /Symposium, which are conducted on topics, related to our work and present papers.
4. To study books, journals and monographs on fixed points for research work.